



Health & Safety

Standard vehicles / pedestrian interaction



Reference

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Details of changes in version C:

- Addition of instructions relating to speed limits
- Additions to the roles of the operational manager and Operators of machinery
- Addition of the requirement for a medical certificate and confirmation that no medication causing drowsiness is being taken in order to issue a driving authorisation
- Addition of the ban on motorized vehicles with 2 or 3 wheels for travel.

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1. Objective

The purpose of this document is to describe the general measures to be taken by Bouygues Travaux Publics for the operation of machines and vehicles in interface with other activities. This involves preventing the risks associated with this type of activity in accordance with the regulations, codes and standards in force in all the countries where BYTP operates.

2. Scope

This document applies to all companies working on sites (including service providers and subcontractors). In the case of Joint ventures (JV), Consortiums or co-contracting operations, the project-specific procedure must include, as a minimum, the provisions set out in this document as well as any non-conflicting provisions of our partners. It is important to have all aspects adopted at the start of the project either prior to or as a minimum at Health and Safety kick-off meetings.

If any part of this document conflicts with a local requirement and is less stringent than that local regulation, then the corresponding local regulation(s) apply.

This document does not apply to interfaces with existing buried or aerial networks, lifting operations and activities near water.

3. Reference documents

- BYCN guidelines vehicles / pedestrians interactions
- BYTP-H&S-REF-2200 – Standard Lines of defence and Life Saving Rules
- BYTP-H&S-REF-2215 – Construction machinery and associated maintenance guidelines
- Policy regarding sensitive equipment BYTP and Subsidiaries

4. Definitions and abbreviations

Term / Abbreviation	Description
	<i>Interactions between pedestrians and vehicles in the following situations:</i>
	• Entrances and exits from the sites, workshops and installations under our responsibility of the following Vehicles: • Concrete mixer trucks, concrete pumps, mobile cranes • Material delivery semi-trailers, waste disposal trucks • Machinery carriers • Commercial vehicles, etc.
Vehicle/pedestrian coactivity	• Earthworks: large mass, sanitation, Roads and Miscellaneous Network, quarries, mines, etc. • Movements within workshops and stores • Movements within production plant areas (voussoirs, concrete, asphalt, etc.) • Movements of construction machinery such as nacelles, telescopic handlers, shovels, tipping motors, etc. • <i>Interactions with the public or private domain:</i> • Entrance and exit of the sites/construction site • Along railway lines or busy roads

Inspection	Examination of equipment to confirm that the equipment has a certificate of conformity and is free from defects.
Construction machinery	All equipment moving autonomously or electrically powered: trucks, vehicles on tires or on tracks, personnel lifting platform, forklift, lifting equipment, earthmoving equipment.
Circulation plan	An evolving traffic plan (possibly integrated into the site installation plan) is prepared, checked and issued by the competent persons before any operation involving site machinery. It covers the entrances/exits of machines and pedestrians, the movement of machines on the various site roads, as well as pedestrian access to the various activity zones and site premises. The traffic plan takes the form of signage and effective separation of flows put in place on site. When activity requires the intervention of machines outside the site roads, the procedures and risk analysis relating to this activity illustrate the access of these machines to the related areas and the means of separation of the flows required for the surrounding pedestrians. In the case of roaming operations, or along public roads, the road marking protocols defined by the public authorities are applied. The traffic plan then defines the accessibility conditions (axle load, wheelbase, gantries, etc.).
Site Installation Plan	The purpose of the plan is to bring together, on a graphic document (printed, 3D, electronic, etc.), information relating to the layout of the site and to give an overall view of the progress of the work. It is drawn up in accordance with the schedule, following the phasing of the activities and, consequently, the total or partial displacement of the site installations before the completion of the work. This can also integrate the traffic plan if all the provisions are specified therein. Depending on the scale of the work, several plans may be necessary to ease the reading of the information related to fences and site access, delivery and storage areas, work areas and roads, location and the movement of site equipment (crane, concrete plant, scaffolding, etc.), site premises (toilets, offices and rest areas, assembly point), safety devices and any other useful information.
Periodic General Inspection	The purpose of the periodic technical checks is to assess the condition of the elements of the installation and of the safety devices, whose deterioration could lead to a danger, in order to determine: • If a repair or exchange is necessary as soon as possible • Or if these safety devices can fulfill their function correctly until the next verification. These regular checks do not only consist of checking the overall proper functioning of an installation, but also of carefully examining its components and its safety devices. They are triggered by the head of the establishment, respecting a regulatory or company schedule.

5. Roles and responsibilities

Each Civil Work Division site implements the requirements of this document, in addition to any requirements that may be necessary to manage the risks specific to its environment. Moreover, the Management team organizes and ensures that inspection visits are carried out to ensure the compliance with the provisions defined in these Guidelines.

Pedestrian - Personnel on foot participating in the activity, working nearby on another activity, entering or leaving their workstation. External personnel (visitors, suppliers, customers, etc.) are also considered pedestrians, and must be accompanied by BYCN or similar personnel. His responsibility is to respect the traffic rules on site, in particular those mentioned at the safety induction and those displayed on site.

Traffic man - The traffic man is a trained employee, responsible for traffic and the maneuver of machinery. He manages the co-activity linked to the maneuver of machinery and vehicles on the site and the management of the entry/exit of delivery trucks as soon as they enter the worksites/sites.

Operational manager - The operational manager is an employee representing the employer and having the authority, the means and the competence to exercise decision-making power for the scope under its authority. He must enforce this Standard and monitor its implementation. He ensures in particular that traffic plans, Site Installation Plans, and exclusion zones are designed and updated taking into account the risk assessment.

Equipment Manager – The Equipment Manager is responsible for orders of purchase or rental of equipment. For example, in the context of this document, he is responsible for the rental of construction machinery for the activities for which he is responsible.

Competent Person - A Competent Person is able to identify existing and foreseeable hazards in the vicinity, or hazardous and/or dangerous working conditions for employees and who is authorized to take prompt corrective action. He takes his skills from a training which allows him to be authorized or certified to carry out these tasks.

Operators of machinery and vehicles – Operators of machinery and vehicles must ensure, before starting any maneuver, that no person is present within their operating area.

6. Training, competencies and authorizations

6.1. Trainings

Machinery driver - All personnel involved in the operation of machinery are subject to a formal training with certification attesting to the ability to drive construction equipment, in order to ensure that they have the knowledge, skills and medical aptitude necessary for driving these vehicles.

When the operation of a machine is not regulated by a standardized type of training, the training must be provided according to specific requirements prepared by a competent person.

Holding a permit or a training certificate does not supersede the need for on-the-job training aimed at conveying the specificities of the locations, the operations to be carried out and potential co-activities.

Each machine operator is subject, according to local requirements and regulations, to a skills assessment (retraining) and effective practice of the machines.

6.2. Competencies

Anyone assigned to the role of traffic marshal must be familiar with:

- Site access procedures and site circulation plan(s)
- Communication methods between drivers and pedestrians, to give instructions as clearly and precisely as possible
- Complex manoeuvres, depending on site configurations
- Machinery blind spots, to be able to position oneself in the correct places and give appropriate instructions
- Traffic management methods, to be able to direct and control the movement of vehicles and pedestrians and thus ensure safe and efficient flows
- Signalling equipment and its use (signage boards, flags, cones, or any other traffic control devices)

6.3. Authorizations and permits

Driving license - The appropriate driving license is required for any use of a company vehicle that requires it (site van, flatbed truck, etc.), depending on local regulations. Holding a driving license does not supersede the need for on-the-job training.

Driving Authorization - A driving authorization is issued by the project manager or his representative, for any machinery driver. This follows an assessment of the theoretical and practical skills of the operators of the machinery concerned, as well as a medical examination certifying their fitness to operate such equipment. Operators must not drive under the influence of medication that may cause drowsiness.

The driving authorization is issued annually and/or at the expiry date of the driving certificates in order to ensure their renewal.

When renting a machinery "with driver", specific information is provided to the driver on the conditions of use and the environment of the activity. It complements the driving authorization issued by the rental company.

A visual means is ideally deployed on site to facilitate the identification of drivers authorized to drive the machineries.

7. Equipment conformity

Certificate of Compliance – All machinery used on a project complies with local Regulations. This conformity is characterized by the provision in the machine of a certificate of conformity issued by the manufacturer himself, thus attesting to his commitment and his responsibility in respecting all the regulatory systems to which may be subject. the device he designed.

Periodic General Inspection - All construction machinery used is uniquely identified, registered and subjected to a Periodic General Inspection at least every 12 months (excluding lifting equipment) and 6 months for lifting platforms, by an organization competent, and subject to local regulations. The point of concerns expressed during the periodic inspections are lifted according to the instructions specified in the report. Those requiring immediate immobilization of the machinery are lifted before any (re)use.

Stand-by - Defective, uninspected or unidentified construction machinery is put on stand-by and is no longer used until it is brought back into conformity. A visual means is put in place on the machinery to remind the ban on its use.

Condition state - All construction machinery is inspected by the operator prior its use.

Maintenance - Protected areas of intervention are defined for all maintenance operations. These appear on the Site Installation Plan.

8. Management of pedestrian and machinery traffic

8.1. Flows separation, parking lots

As far as possible, pedestrian paths and those of vehicles are completely separated to avoid any risk of collision. In the event of proximity, the movement of machineries and vehicles are separated from those of pedestrians by physical separations adapted to the nature of the risk (when not possible, traffic men are designated) in order to ensure that there is no uncontrolled vehicle-pedestrian coactivity. Pathways and worksite tracks are permanently maintained cleaned and cleared.

Pedestrian crossing zones are marked, secured and equipped at intersections with vehicle traffic with restraint devices to avoid crossing without stop (gate type separation).

The speed limit is defined according to customer requirements, site conditions and the risk analysis associated with each site.

Physical protective barriers - The systematic installation of protective barriers around nearby workstations and in interaction with vehicle traffic is mandatory. Nevertheless, in the case of a short-term or roaming operation making materialization difficult, a specific instruction is defined and formalized by the management/operational teams.

Traffic marshal - A traffic man is mandatory for all operations involving:

- Reverse maneuvers without visibility or with limited view (blind spot, etc.) via camera or other detection systems
- When it is impossible to use alternative means (such as cameras, detection system)
- When it is impossible to physically separate vehicle-pedestrian flows,
- If the risk analysis reveals a need for the management of machinery-machinery or machinery-pedestrian flows

The traffic marshal must be identifiable on site (high-visibility vest, coloured helmet, etc.).

The traffic man reports to the relevant area manager of the area or site, or to one of these deputies (site manager).

Parking - All construction machinery is parked at the end of use in a specific location. A safe distance between each vehicle must be respected to allow the movement of drivers and avoid contact between vehicles. If it is technically or organizationally impossible to define suitable parking areas, a specific instruction defines the provisions for parking vehicles. Stationary vehicles switch off the ignition and engage the handbrake. When a driver leaves a vehicle on sloping ground, outside the defined location of its parking or unloading area, chocks are positioned under the wheels to prevent any unintentional movement.

Wheel deflector - The danger zones (slope, hopper, etc.) are materialized by a "wheel deflector" device preventing the approach of nearby vehicles. In the event of technical impossibility, a minimum safety distance is defined and respected according to the activity to be carried out and the motion speed of the machineries.

Briefing - In the case of co-activity between vehicles and pedestrians, a pre-start briefing is systematically carried out by the designated person in charge of the operation, specifying the roles and responsibilities of each person involved in the situation, in particular during modifications to the traffic plan.

8.2. Circulation Flow

Traffic lanes are one-way when possible; if necessary, reversal zones are materialized, as well as any other provision allowing to manage the risk of collision between a pedestrian and a vehicle.

Reverse gears are only permitted if the risk analysis, depending on the specificities of each site, has not made it possible to be avoided. Human and technical resources are then implemented (traffic man, control screens, etc.).

8.3. Rules and illumination of traffic areas

Driving or using any motorised two- or three-wheeled vehicle is prohibited for any work-related travel (business trips, travel to or between construction sites, etc.).

The traffic rules are presented in a visible way by the use of arrows and signs (such as stops, give way, speed limits, parking zones, prohibited access, crossings, pedestrian crossings, vehicle crossings, indication of turns without visibility, steep slopes, road works) and are at least compliant with the signage applied on public roads in the country.

Machinery and pedestrian traffic are equipped with signs indicating the different areas of the site, which allow drivers to easily find their way around the sites.

Safety distance - When there is no separation for traffic (one time operation, temporary operation, etc.), that the driver does not have visual control of nearby people and that they are not directly involved in the driving operation of the vehicle, these people maintain a safety distance of at least 2 meters from the vehicle. This safety distance is materialized by temporary markings.

In accordance with our Life-Saving rule, no one places themselves within the radius of action of a machinery without the activity requiring it and without having been authorized by the driver of the machinery concerned. A traffic man can complete the team in order to ensure coordination and avoid the risk of collisions.

Light and visibility - The areas where the machineries are moving are sufficiently lit (40 lux), in particular the sensitive places (intersection, crossing, other workstations, etc.). The level of illumination of the interior circulations is 40 Lux.

Written rules of use are defined on sites requiring additional means of protection (high visibility vests, LED vests, lighting balloons, etc.).

Speed limit – Depending on the configuration of the site's roads, it is necessary to establish speed limits and to ensure that appropriate speed limit signs are installed.

8.4. Construction machinery safety equipment

For more details on equipment specific to machinery, refer to **BYTP-H&S-REF-2215 (Standard "Construction machinery and associated maintenance")**.

Mandatory equipment and instructions (at a minimum) on construction machinery:

Seatbelt - Wearing a seatbelt is mandatory .

Beacon - A beacon-type signaling device is found on each machine and heavy vehicle . A flashing light type signaling device (of a different color from that mentioned above) or similar is recommended to enable remote identification of the effective wearing of the seat belt.

Reverse beep/Cri du lynx® - A sound device alerts you to any movement in reverse.

Safety device – A visual signaling device for any machine movement such as tipper, jib, fork, stabilizers, sides, auxiliary crane which are not in a secured state.

Keys - The keys are the responsibility of the driver which keep them as soon as he leaves his vehicle. All the keys are recorded at the end of use to the person in charge designated by the related operational, logistics or company manager.

Communication - A method of operational communication is defined beforehand between the exposed personnel and the operator driving the machine. Only the designated responsible person provides instructions to the machine operator. However, the operator obeys a stop signal at all times, regardless of who gives it.

Additional equipment on construction machinery:

Cameras, reversing radar and detection systems - All machines and vehicles (excluding light vehicles) specific to the site and working in areas accessible to pedestrians (workers, technicians, etc.) are equipped with cameras or reversing radar:

- Depending on the type **AND**
- Depending on the worksite risk assessment (coactivity, constrained space, etc.)

If it is impossible to equip them, a specific instruction includes the systematic use of a traffic man for any movement in pedestrian interaction or maneuvering areas.

Depending on the risk analysis carried out considering the specific conditions of the site, the activity to be performed and the planned coactivity, a pedestrian detection system can be implemented in addition to the reversing cameras and/or radar.

8.5. Use of cell phones and radios

Fundamentals:

- On site, any use of the telephone or radio-type equipment is prohibited while driving.
- Outside construction sites, any manipulation of the phone while driving is prohibited.

The use of communication systems (Bluetooth, hands-free kit, etc.) is the subject of a specific instruction.

The use of mobile phones on site by pedestrians in motion is prohibited.

9. Appendice

- Annex 1 – Signaling standards and vehicle-pedestrian protection systems – BYTP-H&S-INF-2168